
Using AI Image Editing Tools for Augmenting 3D Gaussian Splatting

Vedant Agrawal
UMD

vedant99@umd.edu

Yuxuan Bu
UMD

ybu1@umd.edu

Saunak Roy
UMD

saunakr@terpmail.umd.edu

Baozan Yan
UMD

byan1@umd.edu

Niko Zhang
UMD

nzhang10@umd.edu

Abstract

3D Gaussian Splatting (3DGS) enables real-time novel view synthesis with high visual quality, but it often breaks when the input views are sparse, noisy, or taken with limited angles (a common case for mobile users). In this project, we study whether modern AI image editing tools and extra vision priors can help sparse 3DGS. We built and tested three practical pipelines: (1) a closed-loop method that renders a video from a sparse 3DGS model, cleans each frame using a depth-map-guided image generation API, and re-trains with the cleaned frames; (2) a post-processing method that edits 3DGS renders using Qwen Image Edit, then evaluates and optionally re-trains; and (3) a sparse-oriented splatting pipeline (SparseGS) that uses monocular depth cues (DepthAnythingV2) and optional diffusion refinement. Across experiments on Mip-NeRF 360 (bicycle) and NeRF Synthetic (chair), we find that naively adding 2D edited images does not give stable improvements in PSNR/SSIM/LPIPS. The main reason is multi-view inconsistency: single-image editors can improve local textures but may change geometry, lighting, or camera-related cues across views, which creates floaters and hurts reconstruction. Our results suggest that future work should focus on 3D-aware or pose-conditioned generation, and on stronger filtering for multi-view consistency.

1 Introduction

Novel view synthesis aims to render a scene from new camera viewpoints given a set of captured images. Neural Radiance Fields (NeRF) [1] produce strong results but are slow at render time because they need expensive volume rendering. 3D Gaussian Splatting (3DGS) [2] is a newer approach that represents scenes as many anisotropic Gaussians and renders them efficiently through rasterization, achieving real-time performance.

However, standard 3DGS works best when the training views are dense and cover the scene well. In real life, users often provide only a small number of images, with limited angles, motion blur, or low light. With sparse views, the reconstruction becomes under-constrained: geometry can drift, view-dependent appearance is unstable, and “floaters” (floating Gaussians) appear. This project asks a simple question:

Can modern AI image editing tools help sparse 3DGS by cleaning bad renders or synthesizing better training images?

We started from the idea that diffusion-based editing can improve low-quality images, and we tried to feed these improved images back into training. In practice, we found that the hardest part is not making an image look better in one view, but keeping the scene consistent across many views.

2 Related Work

2.1 3D Gaussian Splatting and its Extensions

3DGS [2] optimizes Gaussian parameters to match input images and can render in real time. Many follow-up works improve compression, pruning, and efficiency [3, 4, 5, 6, 7]. A key open issue is robustness under sparse views. Sparse-view settings motivate methods that add priors such as depth and regularization; SparseGS [8] is one example designed for sparse 360-degree view synthesis.

2.2 Diffusion Models and Image Editing

Diffusion models [9] and large multimodal image editors like Qwen Image Edit [10] and Step1X-Edit [11] can perform strong single-image editing, inpainting, and enhancement. Self-supervised vision models like DINOv2 [12] provide robust visual features and can support filtering or retrieval. Yet, most 2D editing models do not enforce multi-view 3D constraints. Recent work explores view-consistent diffusion priors for 3DGS [13] and instruction-based 3D editing [14], which inspires our future directions.

3 Problem Setup

3.1 Goal

Given a sparse set of input images of a scene, our goal is to train a splatting-based renderer and synthesize high-quality images from previously unseen viewpoints. We investigate whether augmenting the training data with AI-edited images, or incorporating additional depth priors, can improve the final reconstruction quality under sparse-view conditions.

3.2 Metrics

We evaluate reconstruction quality using three standard metrics commonly adopted in novel view synthesis. PSNR is used to measure pixel-level fidelity between rendered images and ground-truth views, with higher values indicating better reconstruction accuracy. SSIM assesses structural similarity and is likewise maximized when the reconstructed image preserves spatial structure. LPIPS measures perceptual distance based on deep feature representations, where lower values correspond to higher perceptual similarity.

3.3 Datasets

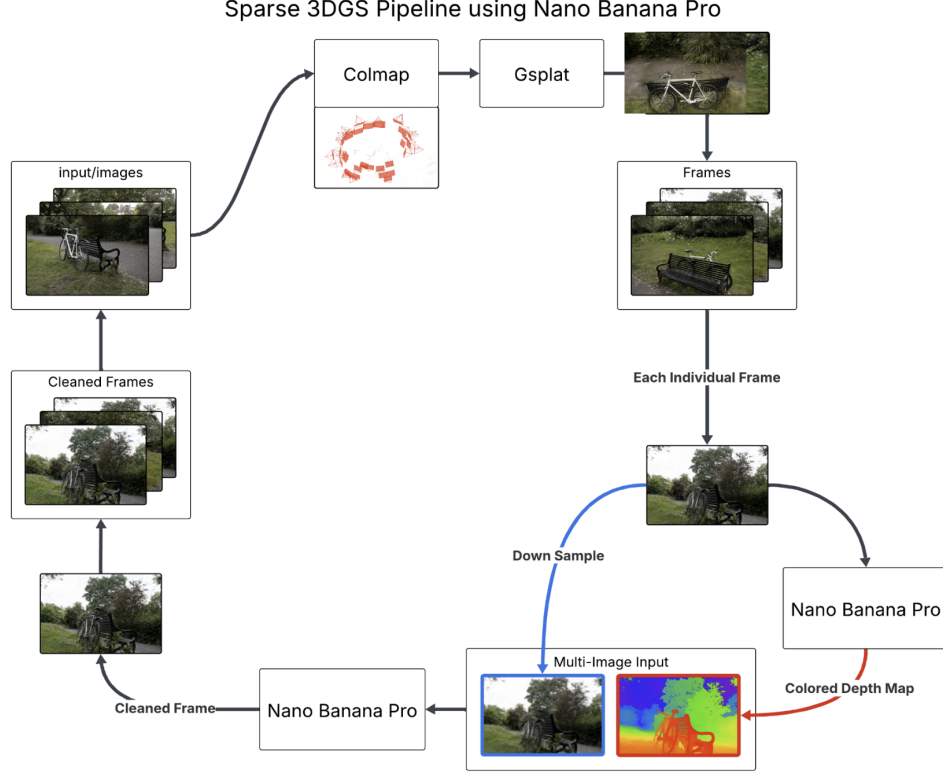
Our experiments are conducted on two widely used datasets that represent complementary evaluation settings. The Mip-NeRF 360 *bicycle* scene provides a real-world environment with complex geometry and appearance variations; to simulate sparse-view scenarios, we subsample the available images to retain only 25% and 15% of the original views. In contrast, the NeRF Synthetic *chair* dataset offers clean ground-truth geometry and appearance, where training and test splits primarily differ in camera viewpoints, enabling more controlled and interpretable evaluation of reconstruction behavior.

4 Methods

We implement three practical pipelines for incorporating AI-based image editing and depth priors into sparse 3D Gaussian Splatting (3DGS). All approaches follow a common structure: a 3DGS model is first trained using sparse input views, additional images or geometric cues are then introduced, and the model is either retrained or evaluated to assess the impact on reconstruction quality. The three pipelines differ in where auxiliary information is injected and how tightly it is coupled with the 3D reconstruction process.

4.1 Approach 1: Closed-loop frame cleaning with depth-map guidance

In the first pipeline, we perform closed-loop refinement by cleaning low-quality rendered frames and feeding them back into training. We begin by running sparse reconstruction with COLMAP



(a) Pipeline

Figure 1: Pipeline map

and discarding clearly inconsistent input images when possible. A 3DGS model is trained using the remaining sparse views, after which a video is rendered along a predefined ellipsoidal camera path and decomposed into individual frames.

Each rendered frame is processed using a two-step image editing procedure. First, a colored depth map is generated for the frame. The downsampled RGB image together with the corresponding depth map is then used to reconstruct a cleaner version of the frame. The cleaned frames are added back into the training set, and the 3DGS model is retrained in a closed-loop manner for a small number of iterations. While depth guidance helps suppress some extreme hallucinations at the single-image level, multi-view inconsistencies remain.

4.2 Approach 2: Post-processing 3DGS renders with Qwen Image Edit

The second pipeline applies image editing as a post-processing step without directly modifying the training procedure. A sparse 3DGS model is trained in Nerfstudio, and a set of novel-view images is rendered using camera poses from the test split. The rendered images are preprocessed to ensure consistent resolution and formatting, then edited using a fixed ComfyUI workflow with Qwen Image Edit.

Edited renders are compared against the corresponding ground-truth test images using PSNR, SSIM, and LPIPS. In additional experiments, the edited images are merged into the original training set and used to retrain the 3DGS model. Because the editing process operates independently on each image, view-dependent changes in appearance or subtle geometric cues may be introduced across views.

4.3 Approach 3: Depth priors with SparseGS

The third pipeline explores a reconstruction method designed explicitly for sparse inputs. We construct extremely sparse training sets, in some cases using as few as 12 views. Monocular depth maps are predicted for each training image using DepthAnythingV2 and are incorporated as additional supervision during training. A SparseGS model is then trained using both RGB images and the predicted depth cues.

Novel views are rendered from the trained model for evaluation. In exploratory experiments, a small subset of rendered images is optionally refined using diffusion-based image editing tools. This approach yields strong reconstructions on training views, but generalization to unseen viewpoints remains challenging under extreme sparsity.

5 Experiments and Results

5.1 Approach 1: Nano Banana Pro closed loop (Mip-NeRF 360 bicycle)

A successful example. Figure 2 shows a case where the cleaned frame looks sharper and less distorted.

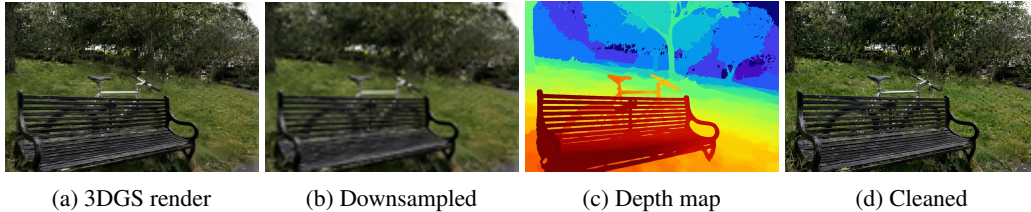


Figure 2: Approach 1 example (25% views). Depth-map-guided cleaning can reduce local distortion in a single view.

A clear failure case (hallucination). Even with depth guidance, the method sometimes changes background content and creates geometry that does not match other views. Figure 3 shows one example.

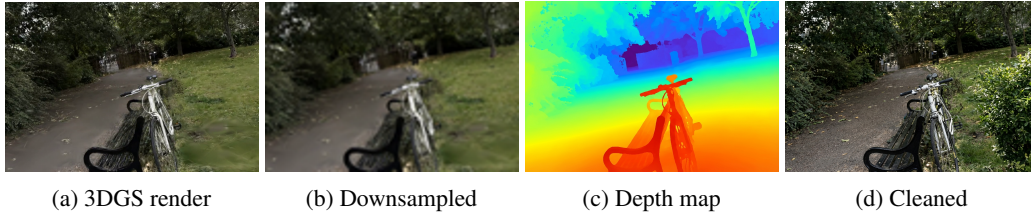


Figure 3: Approach 1 failure case: the cleaned output introduces hallucinated background/structures that can break multi-view consistency.

Extreme sparsity can collapse. When the base 3DGS renders are too poor (15% views), the cleaning model often cannot infer the true scene and drifts to a different one. The drift becomes worse after multiple closed-loop rounds. Figure 4 shows an example.

Quantitative results. Table 1 shows that our closed-loop augmentation did not beat the baseline, and often made SSIM/LPIPS worse.

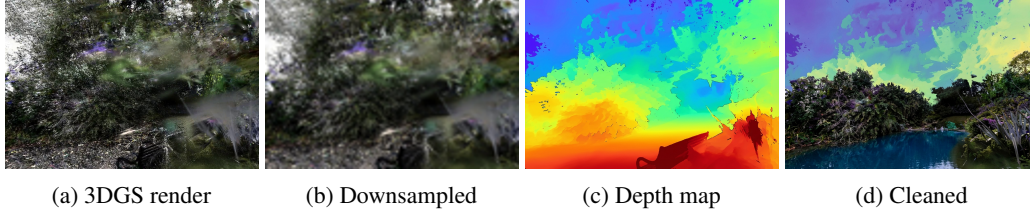


Figure 4: Approach 1 under extreme sparsity (15% views): the model may hallucinate a different scene, and closed-loop re-training accumulates errors.

Table 1: Approach 1 on Mip-NeRF 360 bicycle. Higher is better for PSNR/SSIM; lower is better for LPIPS.

Setting	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
25% views (baseline)	16.98	0.44	0.38
25% views (ours)	16.76	0.35	0.41
15% views (baseline)	12.72	0.31	0.52
15% views (ours)	12.15	0.17	0.61

5.2 Approach 2: Qwen Image Edit post-processing (NeRF Synthetic chair)

We evaluate whether a strong 2D image editor (Qwen Image Edit via ComfyUI) can serve as a practical post-processing tool for sparse 3DGS. Concretely, we first train a baseline 3DGS model on sparse training views, render novel views (or matched camera poses when available), and then apply Qwen-based editing to each rendered frame independently. We report both qualitative observations and quantitative metrics (PSNR/SSIM/LPIPS) against the held-out test images.

Qualitative behavior. Across many frames, Qwen edits tend to improve *single-view* appearance: noise is reduced, textures become smoother, and some local artifacts look less distracting. However, these improvements are often not *structure-preserving* in a multi-view sense. Even when the object identity (e.g., the chair silhouette) remains recognizable, small view-dependent changes can appear in edges, specular patterns, and fine geometry cues (e.g., holes, thin bars, or leg boundaries). Such changes are difficult to detect by eye on a single image, but they can accumulate across views and manifest as floaters or smeared surfaces after re-training.

Quantitative results. Table 2 summarizes the main evaluation on the test set. In our experiments, Qwen-edited renders do not yield consistent improvements over the baseline renders under PSNR/SSIM/LPIPS. This indicates that making a render look cleaner in one view does not reliably reduce the error with respect to ground-truth test images, and the edits may even move the image distribution away from the test targets.

Table 2: Approach 2 main results on NeRF Synthetic chair (test images compared against baseline renders and Qwen-edited renders).

Comparison	Metric	mean	std	median
Test vs Baseline 3DGS	PSNR $\uparrow$	21.6244	7.7138	18.1496
	SSIM $\uparrow$	0.9050	0.0593	0.9097
	LPIPS $\downarrow$	0.1743	0.1151	0.1405
Test vs Qwen-edited	PSNR $\uparrow$	18.3977	4.9964	17.3151
	SSIM $\uparrow$	0.8842	0.0496	0.8839
	LPIPS $\downarrow$	0.2130	0.0961	0.2191

How much does Qwen change the baseline renders? To characterize the magnitude of the edits (without implying they are beneficial), we also compute metrics between baseline renders and Qwen-

edited renders (Table 3). These numbers should be interpreted carefully: a high render-to-render PSNR/SSIM can occur even when the edits introduce small but view-dependent shifts in edges, shading, or fine details. Such shifts may have limited impact on pixel-wise similarity between the two renders, yet still break multi-view geometric consistency and harm 3D reconstruction when used for augmentation.

Table 3: Baseline renders compared against Qwen-edited renders (measuring edit magnitude, not correctness).

Comparison	Metric	mean	std	median
Baseline 3DGS vs Qwen-edited	PSNR $\uparrow$	29.3138	12.6333	24.5813
	SSIM $\uparrow$	0.9428	0.0474	0.9495
	LPIPS $\downarrow$	0.1107	0.0752	0.1085

5.3 Approach 3: SparseGS + DepthAnythingV2 (very sparse setting)

Approach 3 shows a common pattern in very sparse reconstruction: training views look good, but test views are much worse.

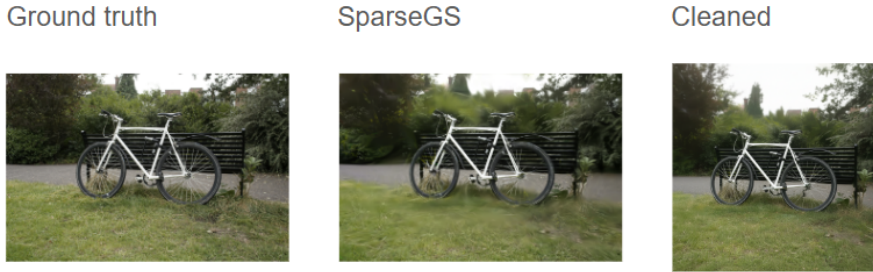


Figure 5: Approach 3: strong reconstruction quality on training views under an extremely sparse (12-view) setting.

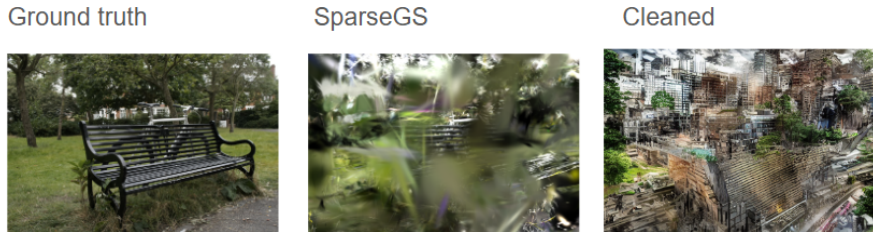


Figure 6: Approach 3: performance drops on held-out test views, with noise and structural drift.

6 Implementation and Reproducibility

Our code is available on GitHub: https://github.com/Baozpy/CMSC473_3D_Gaussian

The repository is organized by pipeline: `pipeline/` implements Approach 1, including dataset sampling, COLMAP-based filtering (when available), and the closed-loop cleaning scripts; `qwen-image-edit/` contains the preprocessing and evaluation scripts used for Approach 2, such as frame processing and camera-transform conversion; and `scripts/` provides helper code for Approach 3, including depth extraction, view subsampling, and exploratory diffusion refinement.

We also included unit tests for several critical utilities, mainly for image resizing/renaming behavior and basic API response handling.

7 Discussion

Across all three approaches, the dominant failure mode can be attributed to the lack of multi-view geometric consistency. Modern 2D diffusion and image editing models, including Qwen Image Edit, are primarily optimized for single-image perceptual plausibility rather than enforcing consistency across multiple viewpoints of a shared 3D scene. Consequently, edits that appear visually reasonable when viewed in isolation often fail to correspond to a coherent underlying geometry when aggregated across views.

In the post-processing setting, this mismatch commonly manifests as background hallucinations or texture alterations that are locally plausible but geometrically inconsistent, leading to view-dependent drift and the emergence of floaters in the reconstructed 3DGS representation. Similar phenomena are observed in data augmentation and closed-loop retraining approaches, where small inconsistencies in camera alignment, object shape, or lighting interpretation introduced by 2D editing are repeatedly reinforced during subsequent training iterations, resulting in compounding errors and progressive degradation of reconstruction quality.

We attempted to mitigate these issues by introducing depth maps as an additional geometric constraint. However, this strategy reveals several inherent limitations. First, reliable depth estimation itself remains challenging in complex scenes, such as those containing thin structures or highly irregular geometry. In our experiments on scenes like *bicycle*, depth maps often fail to capture meaningful geometric variation, producing overly smooth or nearly uniform regions for complex elements such as grass or foliage. Second, it is unclear whether diffusion-based image editing models can effectively interpret depth maps as conditioning signals, as these models are not explicitly trained to reason over abstract geometric representations beyond RGB appearance. Finally, while depth information may help constrain gross geometric or positional inconsistencies, it offers little guidance for enforcing consistency in appearance-related attributes such as color, texture fidelity, or surface completeness. As a result, depth-aware conditioning alone is insufficient to prevent view-dependent appearance drift, which continues to accumulate across views and training iterations.

Taken together, these observations suggest that without explicit multi-view or geometry-aware mechanisms, purely 2D image-based diffusion models are fundamentally misaligned with the requirements of 3D reconstruction pipelines. Their objective functions prioritize perceptual realism at the image level, which does not directly translate into stable, view-consistent structure in 3D scene representations.

8 Future Work

Our results indicate that the primary limitation of the explored approaches lies in the structural mismatch between 2D image-based generative models and the requirements of multi-view consistent 3D reconstruction. Future work should therefore prioritize mechanisms that either explicitly prevent inconsistent 2D edits from influencing the 3D pipeline, or fundamentally align the generative process with geometric constraints.

One practical direction is to introduce explicit selection or rejection mechanisms into the generation pipeline. Instead of assuming that all edited images are beneficial, generated frames could be filtered based on multi-view consistency checks or downstream metrics, ensuring that only edits that preserve geometric stability are retained. Such filtering could serve as a safeguard against error accumulation in closed-loop augmentation settings.

Beyond post hoc filtering, a more principled improvement would involve adapting generative models to individual scenes or objects. Scene-specific fine-tuning may reduce hallucinations by constraining the model’s output space, encouraging artifact removal while discouraging structural modifications. However, this approach alone does not resolve the fundamental lack of geometric awareness.

A more promising long-term direction is to move beyond single-image editing altogether. Generative models that explicitly condition on camera pose or operate jointly over multiple views—such as pose-conditioned view synthesis or trajectory-level diffusion—are better aligned with the requirements of 3D reconstruction. By modeling correlations across viewpoints, these approaches have the potential to enforce cross-view consistency directly, rather than relying on indirect constraints. Together, these

directions suggest that future progress will likely require generative models that are geometry-aware by design, rather than purely optimized for 2D perceptual realism.

9 Conclusion

In this project, we evaluated three practical strategies for incorporating AI-based image editing and depth priors into sparse 3D Gaussian Splatting pipelines. Although post-processed renders occasionally exhibit improved visual quality at the single-image level, these gains do not translate into consistent improvements in reconstruction metrics or overall 3D stability.

Our experiments reveal that the primary bottleneck is multi-view consistency. Image editing and diffusion models are optimized for perceptual plausibility in individual views, but they lack mechanisms to enforce that edits across different viewpoints correspond to a coherent underlying 3D scene. As a result, visually appealing edits can introduce subtle geometric or appearance inconsistencies that accumulate across views, ultimately degrading 3D reconstruction quality.

These findings suggest that improving sparse 3D reconstruction cannot be achieved by image-level refinement alone. Instead, meaningful progress will likely require generative models that are explicitly aware of geometry and camera pose, as well as stronger consistency-driven selection or filtering mechanisms. More broadly, this work highlights the fundamental gap between 2D perceptual optimization and the structural requirements of multi-view 3D reconstruction.

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